

EASTERN COALFIELDS LIMITED & CMPDI RI-I ASANSOL

Rajmahal Coalfield Deep Exploration & Mineral Resource Estimation

RAJMAHAL EXPANSION MASTER EXPLORATION DOSSIER (2026)

Mine Block: Lalmatia & Hura Basin Sector (Seam III Block)

Operating Subsidiary: Eastern Coalfields Limited (ECL)

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1. EXECUTIVE OVERVIEW

Rajmahal Coalfield represents a major power-grade coal deposit in Jharkhand. The basin supplies critical thermal fuel to NTPC Farakka and Kahalgaon Super Thermal Power Stations. This technical dossier establishes proved geological resources across the main mining seams.

CERTIFIED GEOLOGICAL PARAMETERS:

- Total Geological Resource: 1,250.00 Million Tonnes (MT) in Seam III.
- Average Seam Thickness: 14.2 meters.
- Overburden Stripping Ratio: 1.85 m³/tonne.
- Ash Content: 24.5% to 32.0%.
- Gross Calorific Value (GCV): 4,800 kcal/kg (Dominant Grade: G10).
- Moisture Content: 7.2% average.

2. SEAM STRATIGRAPHY & QUARRY BENCH SPECIFICATIONS

Seam III is the principal deposit in the Lalmatia area. The seam occurs at shallow depths (between 30m and 160m), making it ideal for high-capacity dragline and shovel-dumper operations. The overlying overburden consists predominantly of coarse-grained sandstones and siltstones with good bench stability parameters.

RAJMAHAL COALFIELD · SEAM III RESOURCE CLASSIFICATION

Quarry Sector	Seam Name	Thickness (m)	Ash Content (%)	Geological Resource (MT)
North Block Quarry	Seam III Top	8.2 m	26.4%	450.00 MT
Central Sector	Seam III Main	14.2 m	28.5%	520.00 MT
South Dip Extension	Seam III Bottom	6.4 m	31.2%	280.00 MT
CUMULATIVE DEPOSIT	Seam III Combined	14.2 m (Avg)	24.5% - 32.0%	1,250.00 MT